Tanvir Arafin

Department of Cyber Security Engineering
George Mason University, Fairfax, VA 22030

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Academic Degrees

University of Maryland **College Park, MD**
Ph.D., Electrical Engineering *2018*

University of Maryland **College Park, MD**
M.Sc., Electrical Engineering *2016*

Bangladesh University of Engineering & Technology **Dhaka, Bangladesh**
B.Sc., Electrical & Electronic Engineering *2011*

Academic Positions

George Mason University **Fairfax, VA**
Tenure-Track Assistant Professor *2022 – present*
Department of Cyber Security Engineering
Affiliate Faculty
Center of Excellence in C5I

Morgan State University **Baltimore, MD**
Tenure-track Assistant Professor *2019 – 2022*
Electrical and Computer Engineering Department
Assistant Director
Cybersecurity Assurance and Policy (CAP) Center

Bangladesh University of Engineering & Technology **Dhaka, Bangladesh**
Lecturer *2011 – 2012*
Electrical and Electronic Engineering Department

Industrial, Consulting, and Summer Positions

Bloomberg **New York, NY**
Software Engineer *2018 – 2019*

Cyber Innovation Group, Philips **Andover, MA**
Vulnerability Research Intern *2016 – 2017*

Security & Privacy Group, Bosch **Pittsburgh, PA**
Research Intern *2016*

Honors, Awards, & Professional Recognitions

Best Poster/Demo Award
ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec) *2025*

Best Hardware Demo Award: 2nd Place
IEEE International Symposium on Hardware Oriented Security and Trust (HOST) *2024*

Thank-a-Teacher
Stearns Center for Teaching and Learning, George Mason University *2024, 2023*

Featured Paper of the Month
IEEE Transactions on Computers (TC) *2022*

Best Paper Award
IEEE Asian Hardware Oriented Security and Trust Symposium (Asian HOST) *2018*

Best Paper Candidate
ACM Great Lakes Symposium on VLSI (GLSVLSI) *2017*

Goldhaber Travel Grant
University of Maryland, College Park *2015*

A. James Clark School of Engineering Distinguished Graduate Fellowship
University of Maryland, Graduate School *2012*

University Merit Scholarship
Bangladesh University of Engineering and Technology *2011*

Students' Honors and Awards

DAC Young fellow
63rd Design and Automation Conference (DAC),
Ph.D. Student: Yanze Wu *2026*
George Mason University

Best Paper Award
8th Annual Sage Memorial Capstone Design Competition,
Undergraduate Students: Justin Rockwell, Max Bedewi, Mack Bartsch,
Bisesh Acharya, Craig Kimbal *2025*
George Mason University

Daniel R. Bannister PhD Retention Fellowship
College of Engineering & Computing,
Ph.D. Student: Yanze Wu *2024*
George Mason University

First Place
National Transportation Cybersecurity Competition 2024,
Undergraduate Student: Finn Schafer *2024*
George Mason University

Second Place
National Transportation Cybersecurity Competition 2024,
Undergraduate Students: Noor Mohammed, Sonia Shaukat, Caleb Hughston, Kayleigh Batchos *2024*
George Mason University

Third Place
National Transportation Cybersecurity Competition 2024,
Undergraduate Student: Navraj Singh Gill *2024*
George Mason University

Best Paper Award
7th Annual Sage Memorial Capstone Design Competition,
Undergraduate Students: Blake E Todorowski, Harris Laing, Michael Fox, Rohit Engala *2024*
George Mason University

Student Travel Grant Award
2024 IEEE International Conference on Mobility: Operations, Services, and Technologies,
Undergraduate Student: Harris Laing *2024*
George Mason University

Publications

I have authored/co-authored **4** book chapters, **7** journal articles, and **30** conference papers. My work has been published in top-tier venues such as **DAC, DATE, HOST, ICCAD, TVLSI** and **IEEE TC**.

Citation Count (Google Scholar): 674, based on the data collected on March 23, 2026
h-Index: 14, i10-index: 21
Erdős Number: 6

Publications after joining George Mason University (from 2022 onward)

My research group has produced **15+** refereed publications since 2022, including top conference publications at **DAC, DATE, and HOST**, several led by my PhD student.

The names of GMU students under my supervision are underlined.
∗ represents GMU Ph.D. student.
Information regarding the Impact Factors are collected from IEEE Xplore, and the acceptance rate is reported from the conference proceedings.

Book Chapters

[1] Ahmed, Fahim* and **Arafin, Md Tanvir**. 2025. "Attack Detection and Countermeasures at Edge Devices". In: *Smart Cyber-Physical Power Systems*. John Wiley and Sons, Ltd. Chap. 20, pp. 539–553. DOI: 10.1002/9781394191529.ch20.

Articles in Refereed Journals

[1] Wu, Yanze* and **Arafin, Md Tanvir**. 2025. "ArKANe: Accelerating Kolmogorov Arnold Networks on Reconfigurable Spatial Architectures". In: *IEEE Embedded Systems Letters*. **ESL '25**. DOI: 10.1109/LES.2025.3567918. [**Impact Factor = 2**].

[2] Lu, Z., Wang, X., **Arafin, Md Tanvir**, Yang, H., Liu, Z., Zhang, J., and Qu, G. 2024. "An RRAM-Based Computing-in-Memory Architecture and Its Application in Accelerating Transformer Inference". In: *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*. **TVLSI '24** 32.03, pp. 485–496. DOI: `10.1109/TVLSI.2023.3345651`. [**Impact Factor = 3.1**].

[3] Pan, Yuqian, Lu, Zhaojun, Zhang, Haichun, Zhang, Haoming, **Arafin, Md Tanvir**, Liu, Zhenglin, and Qu, Gang. 2023. "ADLPT: Improving 3D NAND Flash Memory Reliability by Adaptive Lifetime Prediction Techniques". In: *IEEE Trans. Computers*. **TC '23** 72.6, pp. 1525–1538. DOI: `10.1109/TC.2022.3214115`. [**Impact Factor = 3.8**].

Articles in Refereed Conference Proceedings

[1] Fan, Andrew, Wu, Yanze*, Han, Harry, and **Arafin, Md Tanvir**. 2026. "GPU Acceleration of the Sum-Check Protocol Over Towers of Binary Fields for Verifiable Computing". In: *Proceedings of the 2026 Design, Automation and Test in Europe Conference*. **DATE '26**. Verona, Italy: IEEE/ACM. [Accepted, **Acceptance Rate = 24.63%**].

[2] Schafer, Finn, Lam, Ryan, Kumar, Siddarth, Tram, Jimmy, and **Arafin, Md Tanvir**. 2026. "LiDAR Spoofing for Compromising SLAM on ROS2 Using Statistical Beam Adjustment and Temporal Modeling". In: *To appear in the Proceedings of the 2026 Hardware Oriented Security and Trust Symposium*. **HOST '26**. Washington DC: IEEE. [Accepted, **Acceptance Rate = 23%**].

[3] Wu, Yanze*, Park, Devin, and **Arafin, Md Tanvir**. 2026. "ATTINA: Spatial Acceleration of Additive Number Theoretic Transform for Energy-Efficient zkSNARKs at the Edge". In: *To appear in the Proceedings of the ACM Design and Automation Conference*. **DAC '26**. Long Beach, CA, USA: ACM. [**Accepted, Acceptance Rate = 22.3%**].

[4] Lebaku, Prathyush Kumar Reddy, Gao, Lu, Zhang, Yunpeng, Li, Zhixia, Liu, Yongxin, and **Arafin, Md Tanvir**. 2025. "Cybersecurity-Focused Anomaly Detection in Connected Autonomous Vehicles Using Machine Learning". In: *International Conference on Transportation and Development 2025*. **ICTD '25**, pp. 566–580. DOI: `10.1061/9780784486191.051`.

[5] Lu, Zelin, Wang, Yitin, Qu, Gang, and **Arafin, Md Tanvir**. 2025. "Auto-profiling Attack on NTT Hardware Implementations". In: *Proceedings of the 2025 Asian Hardware Oriented Security and Trust Symposium*. **AsianHOST '25**. Nanjing, China: IEEE. [**Acceptance Rate = 47.17%**].

[6] Somavarapu, Sreenithya, Chaudhari, Harshita, Aidlaid, Nour El Houda, Adchariyavivit, Nona, Dib, Qais, Hossain, Moinul, Shah, Vijay K, and **Arafin, Md Tanvir**. 2025. "Radio Unit Activity Fingerprinting through Electromagnetic Side-Channel Analysis in O-RAN Networks". In: *18th ACM Conference on Security and Privacy in Wireless and Mobile Networks*. **WiSec '25**, pp. 304–305. DOI: `10.1145/3734477.3736152`. [**Best Poster/Demo Award, 1/14 submissions**].

[7] Wu, Yanze* and **Arafin, Md Tanvir**. 2025. "Energy-Efficient Acceleration of Hash-Based Post-Quantum Cryptographic Schemes on Embedded Spatial Architectures". In: *Proceedings of the 2025 International Conference on Parallel Architectures and Compilation Techniques*. **PACT '25**. Irvine, California, USA: ACM. [**Acceptance Rate = 27.87%**].

[8] Wu, Yanze* and **Arafin, Md Tanvir**. 2025. "SHAFI: Securing Hash-Based Post-Quantum Cryptography from Hardware Fault Injection Attacks". In: *Proceedings of the 2025 Asian Hardware Oriented Security and Trust Symposium*. **AsianHOST '25**. Nanjing, China: IEEE. DOI: `10.1109/AsianHOST68425.2025.11370367`. [**Acceptance Rate = 47.16%**].

[9] Wu, Yanze* and **Arafin, Md Tanvir**. 2025. "THENA: Accelerating Torus Fully Homomorphic Encryption on Energy-Efficient Heterogeneous Architecture". In: *Proceedings of the 43rd IEEE

International Conference on Computer Design. **ICCD '25**. Dallas, TX, USA: IEEE. [**Acceptance Rate = 25.5%**].

[10] Ahmed, Fahim* and **Arafin, Md Tanvir**. 2024. "SANTA: A Spatial Accelerator Design for Efficient Number Theoretic Transform (NTT) on Heterogeneous System-on-Chips". In: *Proceedings of the 2024 Workshop on Attacks and Solutions in Hardware Security*. **CCS-ASHES '24**. Salt Lake City, UT, USA: ACM, pp. 2–10. DOI: 10.1145/3689939.3695781. [**Acceptance Rate = 32.26%**].

[11] Todorowski, Blake E, Fox, Michael Lane, Laing, Harris E, Gaddam, Kirthan, Mian, Anosh, Ferrari, Jair, and **Arafin, Md Tanvir**. 2024. "Poster: Address Resolution Protocol Based Attacks for Multi-Robot Systems". In: *2024 IEEE International Conference on Mobility: Operations, Services, and Technologies*. **MOST '24**. IEEE Computer Society, pp. 281–282. DOI: 10.1109/MOST60774.2024.00041.

[12] Wu, Yanze* and **Arafin, Md Tanvir**. 2024. "Ising Model Processors on a Spatial Computing Architecture". In: *2024 IEEE 67th International Midwest Symposium on Circuits and Systems*. **MWSCAS '24**. IEEE, pp. 1383–1387. DOI: 10.1109/MWSCAS60917.2024.10658926. [**Late Breaking Result**].

Publications prior to joining GMU

Book Chapters

[1] **Arafin, Md Tanvir**, Xu, Qian, and Qu, Gang. 2022. "Voltage Overscaling Techniques for Security Applications". In: *Approximate Computing*. Springer, pp. 287–311. DOI: 10.1007/978-3-030-98347-5_12.

[2] Xu, Qian, **Arafin, Md Tanvir**, and Qu, Gang. 2022. "Approximation on Data Flow Graph Execution for Energy Efficiency". In: *Approximate Computing*. Springer, pp. 207–232. DOI: 10.1007/978-3-030-98347-5_9.

[3] **Arafin, Md Tanvir** and Qu, Gang. 2021. "Hardware-Based Authentication Applications". In: *Authentication of Embedded Devices*. Springer, pp. 145–181. DOI: 10.1007/978-3-030-60769-2_6.

Articles in Refereed Journals

[1] Zhang, Jiliang, Shen, Chaoqun, Su, Haihan, **Arafin, Md Tanvir**, and Qu, Gang. 2022. "Voltage Over-Scaling-Based Lightweight Authentication for IoT Security". In: *IEEE Transactions on Computers*. **TC '22**. DOI: 10.1109/TC.2021.3049543. [**Impact Factor = 3.8, Featured Paper of the Month, February 2022**].

[2] **Arafin, Md Tanvir** and Qu, Gang. 2018. "Memristors for Secret Sharing-Based Lightweight Authentication". In: *IEEE Transactions on Very Large Scale Integration Systems*. **TVLSI '18** 26.12, pp. 2671–2683. DOI: 10.1109/TVLSI.2018.2823714. [**Impact Factor = 3.1**].

[3] Gao, Mingze, Wang, Qian, **Arafin, Md Tanvir**, Lyu, Yongqiang, and Qu, Gang. 2017. "Approximate Computing for Low Power and Security in the Internet of Things". In: *IEEE Computer*. **TC '17** 50.6, pp. 27–34. DOI: 10.1109/MC.2017.176. [**Impact Factor = 3.8**].

[4] **Arafin, Md Tanvir**, Islam, Nazifah, Roy, Sourav, and Islam, Saiful. 2012. "Performance Optimization for Terahertz Quantum Cascade Laser at Higher Temperature Using Genetic Algorithm". In: *Optical and Quantum Electronics*. **OQEL '12** 44.15, pp. 701–715. DOI: 10.1007/s11082-012-9590-z. [**Impact Factor = 4**].

Articles in Refereed Conference Proceedings

[1] **Arafin, Md Tanvir**. 2022. "Computation-in-Memory Accelerators for Secure Graph Database: Opportunities and Challenges". In: *27th IEEE/ACM Asia and South Pacific Design Automation Conference*. **ASP-DAC '22**. IEEE. [**Invited Paper**].

[2] Wang, Shuangbao Paul, **Arafin, Md Tanvir**, Osuagwu, Onyema, and Wandji, Ketchiozo. 2022. "Cyber Threat Analysis using Artificial Intelligence and Machine Learning". In: *IEEE International Conference on Cryptography, Security and Privacy*. **CSP '22**. IEEE.

[3] **Arafin, Md Tanvir** and Kornegay, Kevin. 2021. "Attack Detection and Countermeasures for Autonomous Navigation". In: *2021 55th IEEE Annual Conference on Information Sciences and Systems*. **CISS '21**. IEEE, pp. 1–6. DOI: 10.1109/CISS50987.2021.9400224.

[4] Lu, Zhaojun, **Arafin, Md Tanvir**, and Qu, Gang. 2021. "RIME: A Scalable and Energy-Efficient Processing-In-Memory Architecture for Floating-Point Operations". In: *2021 26th IEEE/ACM Asia and South Pacific Design Automation Conference*. **ASP-DAC '21**. IEEE, pp. 120–125. DOI: 10.1145/3394885.3431524. [**Acceptance Rate = 30%**].

[5] Xu, Qian, **Arafin, Md Tanvir**, and Qu, Gang. 2021. "Security of Neural Networks from Hardware Perspective: A Survey and Beyond". In: *2021 26th IEEE/ACM Asia and South Pacific Design Automation Conference*. **ASP-DAC '21**. IEEE, pp. 449–454. DOI: 10.1145/3394885.3431639. [**Invited Paper**].

[6] **Arafin, Md Tanvir** and Lu, Zhaojun. 2020. "Security Challenges of Processing-in-Memory Systems". In: *Proceedings of the 2020 ACM Great Lakes Symposium on VLSI (GLSVLSI)*. **GLSVLSI '20**, pp. 229–234. DOI: 10.1145/3386263.3411365. [**Invited Paper**].

[7] Gao, Jiabao, Wang, Jian, **Arafin, Md Tanvir**, and Jinmei, Lai. 2020. "FABLE-DTS: Hardware-Software Co-Design of a Fast and Stable Data Transmission System for FPGAs". In: *2020 IEEE 33rd International System-on-Chip Conference*. **SOCC '20**. IEEE. DOI: 10.1109/SOCC49529.2020.9524764. [**Acceptance Rate = 30.1%**].

[8] Xu, Qian, **Arafin, Md Tanvir**, and Qu, Gang. 2020. "MIDAS: Model Inversion Defenses Using an Approximate Memory System". In: *2020 IEEE Asian Hardware Oriented Security and Trust Symposium*. **AsianHOST '20**. IEEE, pp. 1–4. DOI: 10.1109/AsianHOST51057.2020.9358254. [**Acceptance Rate = 27%**].

[9] Yimer, Tsion, **Arafin, Md Tanvir**, and Kornegay, Kevin. 2020. "Securing Industrial Control Systems Using Physical Device Fingerprinting". In: *IEEE International Conference on Internet of Things: Systems, Management and Security*. **IoTSMS '20**. IEEE, pp. 1–6. DOI: 10.1109/IOTSMS52051.2020.9340160.

[10] **Arafin, Md Tanvir**, Shen, Haoting, Tehranipoor, Mark M, and Qu, Gang. 2019. "LPN-based Device Authentication Using Resistive Memory". In: *Proceedings of the 2019 ACM Great Lakes Symposium on VLSI*. **GLSVLSI '19**, pp. 9–14. DOI: 10.1145/3299874.3317970. [**Acceptance Rate = 27%**].

[11] Jain, Shalabh, Wang, Qian, **Arafin, Md Tanvir**, and Guajardo, Jorge. 2018. "Probing Attacks on Physical Layer Key Agreement for Automotive Controller Area Networks". In: *2018 IEEE Asian Hardware Oriented Security and Trust Symposium*. **AsianHOST '18**. IEEE, pp. 7–12. DOI: 10.1109/AsianHOST.2018.8607166. [**Best Paper Award**].

[12] **Arafin, Md Tanvir**, Anand, Dhananjay, and Qu, Gang. 2017. "A Low-Cost GPS Spoofing Detector Design for Internet of Things (IoT) Applications". In: *Proceedings of the 2017 ACM*

Great Lakes Symposium on VLSI 2017. **GLSVLSI '17**, pp. 161–166. DOI: 10.1145/3060403.3060455. [**Best Paper Nominee, Acceptance Rate = 24%**].

[13] **Arafin, Md Tanvir**, Gao, Mingze, and Qu, Gang. 2017. "VOLtA: Voltage Over-Scaling Based Lightweight Authentication for IoT Applications". In: *2017 22nd IEEE/ACM Asia and South Pacific Design Automation Conference*. **ASP-DAC '17**. IEEE, pp. 336–341. DOI: 10.1109/ASPDAC.2017.7858345. [**Acceptance Rate = 30%**].

[14] **Arafin, Md Tanvir**, Stanley, Andrew, and Sharma, Praveen. 2017. "Hardware-based Anti Counterfeiting Techniques for Safeguarding Supply Chain Integrity". In: *2017 IEEE International Symposium on Circuits and Systems*. **ISCAS '17**. IEEE, pp. 1–4. DOI: 10.1109/ISCAS.2017.8050605.

[15] **Arafin, Md Tanvir**, Anand, DM, and Qu, Gang. 2016. "Detecting GNSS Spoofing using a Network of Hardware Oscillators". In: *Proceedings of the 47th Annual Precise Time and Time Interval Systems and Applications Meeting*. **PTTI '16**, pp. 74–79. DOI: 10.33012/2016.13135.

[16] **Arafin, Md Tanvir** and Qu, Gang. 2016. "Secret Sharing and Multi-User Authentication: From Visual Cryptography to RRAM Circuits". In: *Proceedings of the 26th ACM Great Lakes Symposium on VLSI*. **GLSVLSI '16**, pp. 169–174. DOI: 10.1145/2902961.2903039. [**Acceptance Rate = 25%**].

[17] **Arafin, Md Tanvir**, Dunbar, Carson, Qu, Gang, McDonald, N, and Yan, L. 2015. "A Survey on Memristor Modeling and Security Applications". In: *IEEE International Symposium on Quality Electronic Design*. **ISQED '15**. IEEE, pp. 440–447. DOI: 10.1109/ISQED.2015.7085466.

[18] **Arafin, Md Tanvir** and Qu, Gang. 2015. "RRAM Based Lightweight User Authentication". In: *IEEE/ACM international conference on Computer-Aided Design*. **ICCAD '15**. IEEE, pp. 139–145. DOI: 10.1109/ICCAD.2015.7372561. [**Acceptance Rate = 26%**].

Ph.D. Thesis

[1] **Arafin, Md Tanvir**. 2018. "Hardware-Based Authentication for the Internet of Things". PhD thesis. University of Maryland, College Park. DOI: 10.13016/M2HH6C88R.

Patents

[1] **Arafin, Md Tanvir** and Kornegay, Kevin. Oct. 2025. *Attack Detection and Countermeasures for Autonomous Navigation*. US Patent 12,438,890.

[2] Jain, Shalabh, Wang, Qian, **Arafin, Md Tanvir**, and Merchan, Jorge Guajardo. Mar. 2021. *Method to Mitigate Voltage Based Attacks on Key Agreement Over Controller Area Network (CAN)*. US Patent 10,958,680.

[3] Jain, Shalabh, Wang, Qian, **Arafin, Md Tanvir**, and Merchan, Jorge Guajardo. Feb. 2020. *Method to Mitigate Transients Based Attacks on Key Agreement Schemes Over Controller Area Network*. US Patent 10,554,241.

Awarded Grants, Contracts, & Donations

I have received grants and contracts as PI/Co-PI for over **$9.5M**, out of which my share is over **$1.96M**.

At George Mason University
Total Award: $5,852,964
Total Awarded to Mason: $4,672,964
My Share: $1,505,977

[1] National Science Foundation
Project: Collaborative Research: CISE: Crosscutting Small: SaTC: SAGE: Secure Accelerators for Next-Generation Foundation Models,
Role: **PI**
Total Award: $600,000, Award to Mason: $275,000 My Share: $275,000 *2025 – 2028*
Other Investigators: Wenjie Che (Howard University)

[2] National Science Foundation
Project: Collaborative Research: CyberSTAR: CyberTraining for Secure Transportation and Reliable Autonomy,
Role: **Lead PI**
Total Award: $500,000, Award to Mason: $200,000 My Share: $200,000 *2025 – 2028*
Other Investigators: Lu Gao (UH), Qian Wang (UC-Merced)

[3] National Science Foundation
Project: CyberCorps Scholarship for Service: EAGLE: Empowering American Government Leadership in Cybersecurity through Education,
Role: **Co-PI**
Total Award Amount to Mason: $3,952,971, My Share: $790,594 *2025 – 2029*
Other Investigators: Kun Sun (PI, GMU), Paulo Costa (GMU), Jianli Pan (GMU), Peggy Brouse (GMU)

[4] National Science Foundation
Project: Collaborative Research: An Edge-Based Approach to Robust Multi-Robot Systems in Dynamic Environments,
Role: **PI**
Total Award: $600,000, Amount to Mason: $95,000, My Share: $95,000 *2022 – 2026*
Other Investigators: Kewei Sha (UNT), Bin Tang (CSUDH), Lily Ma(CUNY), Pooyan Fazli (ASU)

[5] Virginia Innovation Partnership Authority
Project: CASPER: Cyber-Analog Sensing for Protecting Critical Energy Infrastructure,
Role: **PI**
Total Award Amount to Mason: $100,000, My Share: $50,000 *2026 – 2027*
Other Investigators: Xiaochu Zhang (UVA)

[6] Virginia Innovation Partnership Authority
Project: Securing Chiplet-based Semiconductor Manufacturing from Untrusted Supply Chains,
Role: **PI**
Total Award Amount to Mason: $50,000, My Share: $50,000 *2024 – 2025*

[7] Virginia Innovation Partnership Authority
Fingerprinting Technology for Enhancing 5G/NextG O-RAN Supply Chain Risk Management,
Role: **PI**
Total Award Amount to Mason: $49,993, My Share: $45,383 *2024 – 2025*
Transferred from Vijay Shah (former PI, NCSU)

Prior to Joining George Mason University

Total Award: ***$3,659,625***
My Share: ***$462,500***

[8] Maryland Industrial Partnerships(MIPS)
Project: VISPR: A Verified Instruction Secure Processor,
Role: **PI**, Total Award Amount: $130,000, My Share: $110,000 *2022*
Institution: Morgan State University

[9] National Science Foundation (NSF)
Project: CyberCorps Scholarship for Service (SFS),
Role: **Co-PI**, Total Award Amount: $3,184,625, My Share: $265,000 *2021 – 2022*
Other Investigators: Kevin Kornegay (PI, MSU), Michel Kornegay (MSU),
Onyema Osuagwu (MSU), Ketchiozo Wandji (MSU)
Institution: Morgan State University

[10] Applied Research Laboratory for Intelligence and Security (ARLIS)
Project: Cyber-Assessment of AI/ML Tools,
Role: **Co-PI**, Total Award Amount: $150,000, My Share: $37,500 *2020 – 2021*
Other Investigators: Kevin Kornegay (PI, MSU)
Institution: Morgan State University

[11] NCAE-C Cyber Curriculum and Research Program
Project: Secure Autonomous Navigation Under Adversarial Attacks,
Role: **Co-PI**, Total Award Amount:$150,000, My Share: $50,000 *2020 – 2021*
Other Investigators: Kevin Kornegay (PI, MSU)
Institution: Morgan State University

[12] NASA Jet Propulsion Lab (NASA-JPL)
Project: Specification-based Anomaly Detection for Embedded Devices,
Role: **Co-PI**, Total Award Amount: $45,000, My Share: $0 *2020*
Other Investigators: Kevin Kornegay (PI, MSU)
Institution: Morgan State University

Invited Talks, Workshops, & Presentations

1. **Hardware for Secure Autonomy**, *C5I Center Research Review*, George Mason University, 2026.
2. **Blueprint for the T-1000: The Next Generation of Foundation Models at the Edge**, *Virginia Technology Summit*, Micron, 2026.
3. **Hardware for Secure Autonomy**, *Innovation in Cybersecurity AI Colloquium Talk Series*, University of the District of Columbia, 2025.

4. **Ghosts in the Shell: Security Issues in Robotic Operating System**, Hardware Demonstration, *Whiskey and Widgets event for College of Engineering and Computing*, George Mason University, 2024.
5. **Hardware Security @ C5I Center**, Hardware Demonstration, *Center Connect event for College of Engineering and Computing*, George Mason University, 2024.
6. **Cyber Security Issues in Modern Robotics**, *EdgeRobot Research Seminar* , California State University - Dominguez Hills, 2023.
7. **Hardware for Secure Autonomy**, *EdgeRobot Summer Research Seminar*, California State University - Dominguez Hills, 2022.
8. **Design of Secure and Efficient Processing-In-Memory Systems for Large-Scale Applications**, Tutorial Presentation, *34th International System-on-Chip Conference (SOCC)*, 2021.
9. **Hardware Lottery and the Perils of Computer Security**, Invited Talk, Computer Science Department, IT University of Copenhagen, Denmark, 2021.
10. **Autonomous Navigation Under Adversarial Attack**, Abstract Presentation, *49th Annual IEEE Applied Imagery Pattern Recognition (AIPR) Workshop*, 2020.
11. **Physical Unclonable Functions for Security Applications**, Invited Talk, COSC Colloquium Series, Computer Science Department, Morgan State University, 2020.
12. **Guided Reinforcement Learning and Imitation Learning: GRILL-SPICE**, (with Terry Stewart) Telluride Neuromorphic Workshop, 2020.
13. **Hardware Security for IoT Devices**, Amazon Graduate Research Symposium, Seattle, Washington, 2017.
14. **Security Data Science: Improving Security with Big Data Techniques**, (with Tudor Dumitras), Maryland Cybersecurity Center(MC2) Annual Symposium, 2014.

Teaching

Courses Taught

George Mason University

- CYSE 650: Adversarial Machine Leraning and Cyber Security *SP 2026*
- CYSE 499/580: Hardware and Cyber-Physical Systems *SP 2023*
- CYSE 465: Transportation System Design *SP 2026, SP 2025, F 2024, F 2023, F 2022, SP 2024*
- CYSE 211: Operating Systems and Lab *F 2024*

Morgan State University

- EEGR 760: Advanced Topics in Computer Engineering *SP 2020*
- EEGR 745: Advanced Digital VLSI Design *F 2021*
- EEGR 480: Introduction to Cyber Security *F 2019, F 2020*
- EEGR 463: Digital Electronics *SP 2022, SP 2021, F 2020, SP 2020, F 2019*

Bangladesh University of Engineering & Technology

- Introduction to Electrical Engineering *SP 2012*
- VLSI I Laboratory *SP 2012, F 2011*
- Microprocessor & Interfacing Laboratory *F 2011*
- Electronics Laboratory *F 2011*

Mentoring & Advisement

Ph.D. Thesis Advisement

George Mason University

- Yanze Wu
- Suhrid Ghosh
- Wangxinlei Chen

Undergraduate Senior Design Project Sponsor

George Mason University

- Project: Fingerprinting 5G Radios for Security Application
  Students: Nour El Houda Aidlaid, Nongnapat Adchariyavivit, Harshita Chaudhari, Qais Dib, Sreenithya Somavarapu *2025*
- Project: GleNN: Large Language Models for Static Reverse Engineering
  Students: Justin Rockwell, Maximilian Bedewi, Craig Kimball, Mack Bartsch, Bisesh Acharya *2025*
- Project: LOCO: Secure Localization for Multi-Robot Systems in Dynamic Environments
  Students: Leigh Black, Shreyas Gadiraju, Andrew Le, Logan Minto, Rahul Patel, Biraj Joshi *2025*
- Project: Securing Robust Multi-Robot Systems in Dynamic Environments
  Students: Blake E Todorowski; Michael Lane Fox; Harris E Laing; Kirthan Gaddam; Anosh A Mian *2024*
- Project: Émpistos: Trusted Embedded Processing and Optimizations for Edge AI Security
  Students: Zack Francis Wagner; AlRaheeq Al Rawas; Anay Gulati; Maximilian A Karen; Bradford Williams *2024*

Doctoral Thesis Committee Member

Morgan State University

- Latha Suravasi, Khir Henderson, Greig Richmond, Edmund Smith, Tsion Yimer

Undergraduate Senior Design Project Supervisor

Morgan State University

- Maryline Ivana Happy; Jose Dominguez-Cortez; Robert Hill *2022*
- Ashia Mccalla; Gerald Amory *2022*
- Antwaan Thomas; Faizat Kaffo *2021*
- Malik Smith; Anthony Turner *2021*
- Fitsum Tadasse; Reuben Macintosh *2021*

Professional Service

Grant Review Committee Member

Panelist, Secure and Trustworthy Cyberspace (SaTC), National Science Foundation (NSF) *2022 – 2024*
Propsoal Reviewer, Commonwealth Cyber Initiative (CCI) *2025*

Report Reviewer, USDoT National Center for Transportation Cybersecurity and Resiliency *2025*
Technical Reviewer, Maryland Industrial Partnerships (MIPS) *2021*

Conference Organizing Committee Member

TPC Chair, Section 2: Hardware Security, IEEE Design and Automation Conference (DAC) *2025*
Publication Chair, IEEE Asian Hardware Oriented Security and Trust Symposium *2021 – 2024*
Publicity Chair, IEEE Asian Hardware Oriented Security and Trust Symposium *2025*

Conference Technical Program Committee Member

IEEE Design and Automation Conference (DAC) *2023 – 2024, 2026*
IEEE International Symposium on Hardware Oriented Security and Trust (HOST) *2026*
ACM Great Lakes Symposium on VLSI (GLSVLSI) *2026*
IEEE Asia and South Pacific Design Automation Conference (ASP-DAC) *2021 – 2023*
IEEE International System-on-Chip Conference (SOCC) *2020 – 2021, 2023 – 2024*
IEEE Asian Hardware Oriented Security and Trust Symposium (AsianHOST) *2021 – 2024*

Conference Session Chair

IEEE Asia and South Pacific Design Automation Conference (ASP-DAC) *2020, 2022*
IEEE International System-on-Chip Conference (SOCC) *2020, 2021*
IEEE Asian Hardware Oriented Security and Trust Symposium (AsianHOST) *2021*

Journal Reviewer

IEEE Transactions on Computer-Aided Design of ICs and Systems (TCAD)
IEEE Transactions on Very Large Scale Integration Systems (TVLSI)
IEEE Transactions on Circuits and Systems I (TCAS I)
IEEE Transactions on Information Forensics and Security (TIFS)
IEEE Journal on Emerging and Selected Topics in Circuits and Systems (TETC)
IEEE Internet of Things Journal (IoTJ)
IEEE Network Magazine
ACM Journal on Emerging Technologies in Computing Systems (JETC)
Elsevier Integration, the VLSI Journal
Elsevier Computer & Security
Springer Journal of Hardware and Systems Security

Departmental Service

George Mason University
Member, Departmental Tenure-Track Hiring Committee, CYSE Department *2025 – 2026*
Member, Ph.D. Proposal Committee , CYSE Department *2025 – 2026*
Member, Departmental Tenure-Track Hiring Committee, CYSE Department *2024 – 2025*
Chair, Ph.D. Proposal Committee , CYSE Department *2023 – 2024*
Member, Department Handbook Committee , CYSE Department *2023 – 2024*
Member, Grievance Committee, CYSE Department *2023 – 2024*
Chair, Graduate Committee, CYSE Department *SP 2023*
Member, Advertisement Committee, CYSE Department *SP 2023*
Member, Departmental Tenure-Track Hiring Committee, CYSE Department *2022 – 2023*

Member, Departmental Term-Track Hiring Committee, CYSE Department *2022 – 2023*
Chair, Colloquium Committee, CYSE Department *F 2022*
Member, Graduate Committee, CYSE Department *F 2022*

Morgan State University
Graduate Coordinator, ECE Department, Morgan State University *2020 – 2022*
Undergraduate Coordinator, ECE Department, Morgan State University *2019 – 2020*
Member, Curriculum Development Committee, Ph.D. in Secure Embedded Systems *2020*
Member, Faculty Development Committee,ECE Department, *2019 – 2022*
Member, Cyber Defense Education (CAE-CDE) Re-designation Committee *2020, 2021*
USENIX Campus Representative, Morgan State University *2020 – 2022*

Affiliation

Member, Institute of Electrical and Electronics Engineers (IEEE) *2008 – present*
Member, Association for Computing Machinery (ACM) *2025 – present*
Member, ACM Emerging Interest Group on Trustworthy and Responsible Systems *2026 – present*
Member, USENIX: The Advanced Computing Systems Association *2020 – 2022*
Member, Sigma Xi, the Scientific Honorary Society *2019 – 2023*
Student Member, IEEE Communication Society *2010 – 2015*

In the Media

1. *Digital drive: Protecting the future of transportation*, CEC Courier, 2025.
2. *Cyber sweep: George Mason students take top three spots in national transportation cybersecurity competition*, The George, 2025.
3. *Cybersecurity students prepare for the inaugural DistrictCon Hacker Conference*, The George, 2025.
4. *Arafin conducting research aimed at securing chiplet-based semiconductor manufacturing from untrusted supply chains*, Chiplet Marketplace, 2024.